Evaluating AI for ECG Signal Quality

A practical lesson for Data Science and Biomedical Engineering

Student handout and assessment brief

Format: 150-minute workshop plus 4–6 hours of independent practical work.
Level: Year 1; basic probability, descriptive statistics, and introductory machine learning.
Team size: 2–4 students; mixed-discipline teams are encouraged.
Due date and submission portal: As announced by your instructor.

1. Purpose and learning outcomes

An acquisition team wants to flag noisy ECG segments for review. Your task is to evaluate the evidence for choosing a model, explain its limitations, and defend a reproducible decision. The target is *signal quality*, not disease diagnosis.

By the end of this lesson, you should be able to:

1. Define the prediction target, observation unit, and intended deployment population.
2. Audit feature availability and distinguish segment-level from participant-level evaluation.
3. Calculate and interpret classification metrics and review workload.
4. Develop a reproducible baseline and one candidate model using separate training, validation, and test data.
5. Use AI or peer critique as an input to reasoning, and verify the resulting claims.
6. Defend a decision under an explicit operating constraint.

Learning cycle

Independent attempt → AI or peer critique → verification → explanation. Keep your initial reasoning before consulting an AI tool. Assessment rewards evidence, understanding, and reproducibility; sophisticated algorithms are not required.

Workshop schedule

Minutes	Activity
0–10	Individual opening questions; no AI.
10–30	Core concepts and data orientation.
30–50	Exercise 1: audit a proposed evaluation pipeline; no AI.
50–70	Exercise 2: manual model comparison; no AI.
70–90	Exercise 3: critique and verification with AI or a peer.
90–115	Practical planning and discipline-specific exercise.
115–135	Oral defense and a changed operating constraint.
135–150	Individual exit quiz; no AI.

Opening questions

What exactly should the model predict? Can two distinct segments come from the same person? What would "works on a new participant" mean? What evidence would you need beyond a high accuracy score?

2. Concepts and data

Scenario

The practical dataset contains 2,000 simulated feature records: 50 participants, each contributing 40 nominal 10-second segments. A label of 1 means noisy/review requested; a label of 0 means acceptable. The deployment question concerns previously unseen participants using the same simulated device categories.

Scope of the data. Features were generated directly, with participant effects, device offsets, overlapping distributions, and stochastic labels. No patient data or raw ECG waveforms are included. Labels are simulated judgments, not clinically validated ground truth. Do not claim diagnostic validity, filtering performance, or clinical thresholds from this dataset.

Provided files

File	Contents
`ecg_quality_features.jsonl`	One JSON object per segment; features, identifiers, and target.
`participant_split.json`	Prescribed assignment of participants to splits.
`data_dictionary.md`	Definitions, units, availability, and limitations.
`starter.py`	Data loader only; assessed analysis is left to you.
`submission_checklist.md`	Deliverable checklist and output schemas.

The prescribed split contains 30 training participants (1,200 segments), 10 validation participants (400 segments), and 10 test participants (400 segments). Keep it fixed for the primary analysis. Inspect the dictionary before deciding which columns are eligible predictors.

Evaluation principles

Training data support fitting model parameters and learned preprocessing. Validation data support model and threshold selection. Test data support final evaluation after choices are frozen. A preprocessing transformation that learns means, scales, or imputation values must be fitted on training data only.

Multiple segments from one participant may be dependent. The unit used for splitting and uncertainty estimation must reflect the intended generalization question. A large segment count does not automatically imply a large number of independent participants.

Procedural holdout: Test labels are visible to make the exercise self-contained. Do not inspect test outcomes during feature selection, tuning, or threshold selection. Write a timestamped freeze record before final evaluation. This is not a secure hidden benchmark.

Confusion matrix and metrics

The positive class is **noisy**. TP denotes a correctly flagged noisy segment; FP is an acceptable segment flagged for review; FN is a missed noisy segment; TN is a correctly accepted segment.

$$\text{Accuracy} = \frac{TP + TN}{N}, \qquad \text{Precision} = \frac{TP}{TP + FP},$$

$$\text{Recall} = \frac{TP}{TP + FN}, \qquad \text{Specificity} = \frac{TN}{TN + FP},$$

$$F_1 = \frac{2TP}{2TP + FP + FN}, \qquad \text{Flag rate} = \frac{TP + FP}{N}.$$

For this exercise only, use $C = 5FN + FP$ and $C_{100} = 100C/N$. These are hypothetical cost units. If a metric has a zero denominator, report it as undefined (NA), rather than silently assigning a value.

3. Independent exercises

Exercise 1 — Audit a proposed pipeline (20 minutes)

A development team proposes the following workflow:

1. Standardize features using all 2,000 segments.
2. Randomly allocate 80% of segments to training and 20% to testing.
3. Include participant identifiers and post-review notes among the predictors.
4. Try 20 configurations and select the one with the highest test accuracy.
5. Report this score as performance on previously unseen participants.

Identify at least four concerns. For each, state what information is used, why this matters, and what evaluation claim it affects. Propose a revised workflow of 5–7 steps. Save your reasoning before using AI.

Exercise 2 — Compare two models (20 minutes)

The following counts describe a **separate illustrative test set of 100 segments**. They are not results from the 2,000-row practical dataset and must not be presented as your trained model results.

Outcome	Model A	Model B
TP	10	18
FP	0	18
FN	10	2
TN	80	62

1. Calculate accuracy, precision, recall, specificity, and F_1 for each model.
2. Determine the number of flagged segments and the hypothetical cost C.
3. Explain whether selecting by accuracy alone meets the operational objective.
4. Write an initial recommendation of at most 80 words, stating assumptions and limitations.
5. Identify at least two conclusions that the table alone cannot support.

Workspace: Show formulas, intermediate calculations, and units in your notebook or worksheet. Preserve this attempt as evidence of independent reasoning.

Changed operating constraint

The acquisition team can review at most 20 segments per 100 received. Revisit your recommendation. Discuss whether a new threshold could be investigated, what data would be needed, and which split should support that investigation. Do not invent performance at unobserved thresholds.

Prepare for oral defense

Each member must be ready to explain one metric, one feature decision, and one limitation. Your instructor may change the review capacity or cost assumption and ask you to explain the consequences without AI assistance.

4. Critique, verification, and the practical analysis

Exercise 3 — AI or peer critique (20 minutes)

After saving your initial decision, you may use this prompt:

> Act as a critical reviewer of our model evaluation. The positive class is noisy. Model A has TP=10, FP=0, FN=10, TN=80. Model B has TP=18, FP=18, FN=2, TN=62. Our initial recommendation is: [insert your text]. Identify assumptions, missing evidence, and possible weaknesses. Separate what can be calculated from what cannot be concluded. Do not invent data.

For at least three claims, record: the claim; accepted, rejected, or uncertain status; your verification method; and any change to your reasoning. A correct response can still contain assumptions requiring discussion. You do not need to manufacture an AI error.

If AI is unavailable or you choose not to use it, exchange critiques with another team and use the same verification table. No paid AI account is required.

Practical assignment — Complete after the workshop

1. **Audit the data.** Check IDs, data types, missing values, value ranges, duplicate segments, class balance, and participant separation. Summarize training and validation data before fitting. Reserve test outcome analysis until the final stage.
2. **Audit feature eligibility.** For each field, state whether it exists at prediction time, whether you include it, and why. Do not treat the target as an input. Discuss identifiers, constant fields, and post-review information explicitly.
3. **Build a baseline and a candidate.** Use a simple training-derived baseline, such as a class-prior predictor, and one interpretable candidate such as logistic regression or a shallow decision tree. Explain your selection. Fit learned transformations on training data only.
4. **Select using validation data.** Report the candidate at threshold 0.5 and investigate other thresholds. Select an operating threshold using a stated validation-only rule, considering C_{100} and a flag-rate limit of 0.20. Specify tie handling. A threshold meeting a validation constraint is not guaranteed to meet it on test data.
5. **Freeze the method.** Record features, preprocessing, model configuration, seed(s), and chosen threshold before inspecting test performance. Never retune because of the test result.
6. **Evaluate once.** Evaluate the baseline and frozen candidate on the test set. Report confusion counts, all listed metrics, flag rate, and cost per 100. Explain whether the test workload meets the constraint. If it does not, report this honestly and propose a future validation study.
7. **Analyze limitations.** Compare final performance by device, including subgroup sample counts. Explain why this is not evidence of performance on a new device. Include a confusion-matrix figure and a validation threshold-versus-recall/flag-rate figure.
8. **Complete your discipline-specific task** on the next page. Mixed teams select one primary track and include a short critique from the other perspective.

5. Discipline-specific tasks and individual assessment

Track A — Data Science

Develop a reproducible evaluation design and address the following:

1. Explain why participant-level separation matches the stated deployment question.
2. Estimate a 95% uncertainty interval for final test recall using a participant-level bootstrap with at least 1,000 replicates. Keep the fitted model and threshold fixed. Resample test participants with replacement and retain all their segments, preserving repeated selections. State how you handle replicates with no positive examples.
3. Explain what this interval represents and what it omits, including variation from retraining and the limited number of test participants.
4. Explain why one confusion matrix cannot reconstruct a full ROC curve. State what additional predictions are needed.

An optional row-split comparison must be clearly labeled as a secondary experiment. It must not replace or influence the primary frozen evaluation.

Track B — Biomedical Engineering

Develop a one-page acquisition and review workflow supported by your analysis:

1. Propose three possible sources of degraded ECG quality and describe a measurement or controlled experiment that could investigate each one.
2. Specify acquisition metadata you would collect in a real study and explain why.
3. Propose a quality-labeling protocol, including independent reviewers, an operational label definition, and disagreement resolution.
4. Describe what happens after a segment is flagged, taking review capacity into account.
5. Explain why an unusual waveform is not automatically noise and what information this feature-only dataset lacks.
6. Propose a validation plan for a different device or electrode configuration. Distinguish proposed experiments from experiments actually performed.

Individual exit quiz — 15 minutes, no AI

1. Explain why precision of 100% does not imply that no noisy segments are missed.
2. Explain why the participant is the grouping unit in this assignment.
3. For TP=12, FP=8, FN=3, TN=77, calculate precision and recall.
4. Describe one AI or peer claim you checked and the evidence used.
5. State one conclusion that the illustrative Model A/B table does not justify.

Individual reflection — Submit separately

Write 300–400 words addressing your initial understanding, a checked claim, a decision you can defend, your own contribution, and one unresolved limitation. Use concrete examples from your work rather than general comments about AI.

Academic practice

AI is permitted for critique, debugging, and explanation during the authorized stages. You remain responsible for calculations, code, citations, and interpretations. Log relevant interactions and

verification. Do not upload private or identifiable data. Independent exercises, the exit quiz, and the oral defense must be completed without AI.

6. Required submission and marking

Submit one `ECGQ_TeamXX.zip` per team. Include:

1. `report.pdf`: 6–8 English pages, excluding references and the AI-log appendix. Include problem definition, data audit, methods, results, operating decision, track task, limitations, and member contributions. Keep the manual exercise separate from empirical model results.
2. `analysis.ipynb` with saved outputs or `analysis.py`; equivalent reproducible R code is acceptable.
3. `requirements.txt` with actual package versions, or an equivalent environment specification.
4. `README.md`: setup, one execution command, seeds, relative paths, and expected outputs. The original provided data will be placed beside your entry-point script.
5. `feature_audit.csv`: one row per input field, with availability, inclusion decision, and reason.
6. `test_predictions.csv`: one row for each of the 400 prescribed test segments, from the selected final model.
7. `metrics.csv`: baseline and final model results for validation and test.
8. `ai_log.md`, `initial_decision.md`, and `freeze_record.md`.

Each student separately submits `StudentID_reflection.pdf`. The instructor announces the deadline and LMS destination.

Machine-readable output schemas

Feature audit: `feature,available_at_prediction,include,reason`.
Predictions: `segment_id,participant_id,y_true,p_noisy,y_pred,threshold`. Probabilities must lie in $[0, 1]$; use $\hat{y} = 1$ when `p_noisy` $\geq$ threshold.
Metrics: `model,split,threshold,n,tp,fp,fn,tn,accuracy,precision,recall, specificity,f1,flag_rate,cost_p`
Use fractions for rates, not percentages. Use NA for undefined ratios and explain them.

Marking rubric

Component	Weight
Data audit and evaluation integrity	20%
Calculations, modeling, and interpretation	20%
Critique and verification evidence	15%
Discipline-specific task	15%
Individual oral defense and reflection	15%
Individual exit quiz	15%

The first four components assess team work; the final two assess individual understanding. A higher predictive score does not automatically earn a higher mark. Unsupported claims, unreproducible results, and inappropriate test-set use reduce credit in the relevant components.

Final checks

Rerun in a clean session. Match prediction IDs to the prescribed test segments exactly. Check that confusion counts sum to the evaluation sample size. Ensure every figure has labeled axes and every conclusion has supporting evidence. Retain your original attempt and disclose AI assistance or state "No AI used."

Further reading: UNESCO, *Guidance for Generative AI in Education and Research* (2023), official guidance. This reading supports discussion of responsible AI-assisted learning; it is not the source of the synthetic data.